

Pediatric Septic Shock



Section I: Scenario Demographics

Scenario Title:	Pediatric Septic Shock		
Date of Development:	09/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To allow learners to become more comfortable managing common issues associated with pediatric resuscitation.
CRM Objectives:	1) Communicate effectively with team regarding orders and drug doses. 2) Allocate resources appropriately to manage a distraught parent.
Medical Objectives:	1) Recognize the need for early IO access in critically unwell child where iv unsuccessful. 2) Initiate appropriate investigations and treatment for septic child. Specifically: a. Check capillary blood glucose. b. Administer IV antibiotics. c. Prioritize IV fluid pushes then vasopressors 3) Recognize the need to intubate a septic child with altered LOC.

Case Summary: Brief Summary of Case Progression and Major Events

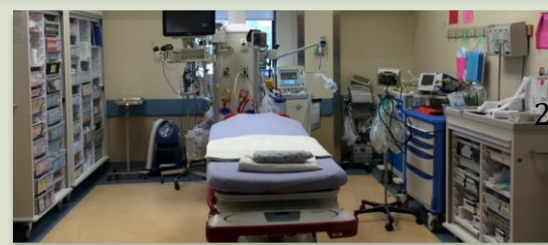
A 4 year-old girl is brought to the ED because she is "not herself." She has had 3 days of fever and cough and is previously healthy. She looks toxic on arrival with delayed capillary refill, a glazed stare, tachypnea and tachycardia. The team will be unable to obtain IV access and will need to insert an IO. Once they have access, they will need to resuscitate by pushing fluids. If they do not, the patient's BP will drop. If a cap sugar is not checked, the patient will seize. The patient will remain listless after fluid resuscitation and will require intubation.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
http://circ.ahajournals.org/content/132/18_suppl_2/S526
http://www.rch.org.au/clinicalguide/guideline_index/Intraosseous_access/



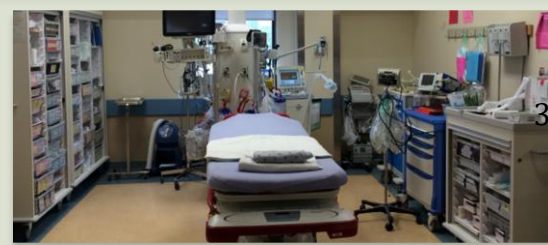
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Pediatric Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other: masks, gowns, gloves for droplet precautions	
☒ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 4-year-old girl presents to your pediatric ED. Her mother states she is “not herself” and seems “lethargic.” She’s had a fever and a cough for the last three days. Today she just seems different. She was brought straight into a resus room and the charge nurse came to find you to tell you the child looks unwell.

B. Patient Profile and History

Patient Name: Rebecca Smythe	Age: 4	Weight: 20kg
Gender: ☐ M ☒ F	Code Status: Full	
Chief Complaint: Lethargic		
History of Presenting Illness: Fever and cough for last three days. Today, not as responsive. Doesn't seem interested in anything. Won't eat or drink. Doesn't look like herself. No known sick contacts , but she does go to pre-kindergarten.		
Past Medical History:	Healthy IUTD Term delivery, no issues.	Medications: None

Allergies: None.

Social History: Lives with mom and dad. Goes to pre-kindergarten class. Has a one year old brother.

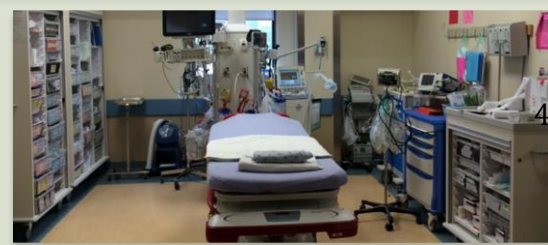
Family History: Dad has asthma.

Review of Systems:	CNS:	Lethargic today. Sort of listless and uninterested.		
	HEENT:	Nil.		
	CVS:	Nil.		
	RESP:	Cough for last three days.		
	GI:	Nil.		
	GU:	Mom doesn't think she's peed today.		
	MSK:	Nil.	INT:	No rashes.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 140/min	BP: 82/44	RR: 40/min
O ₂ SAT: 91%		
Rhythm: Sinus tach	T: 39°C	Glucose: 2.4 mmol/L
General Status: Looks toxic and unwell.		
CNS:	Glazed stare. Lethargic. PERLA.	
HEENT:	Normal TMs. PERLA. Glazed stare.	
CVS:	No murmur. Cap refill 5 seconds centrally. Eyes sunken.	
RESP:	GAEB. Rhonchi to R.	
ABDO:	Nil.	
GU:	Nil.	
MSK:	No hot joints.	SKIN: No rashes.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 140/min BP: 82/44 RR: 40/min O ₂ SAT: 91% T: 39°C	Looks unwell. Cap refill 5 sec. Glazed stare.	<u>Learner Actions</u> ☐ Attempt IV access (unable) ☐ Attempt IO access ☐ Monitors ☐ Apply O ₂ ☐ Septic lab workup ☐ CXR ☐ Push fluids 20ml/kg x3 (400ml per bolus) ☐ Check glucose (2.4) , replace with 2-4ml/kg of D25 (40-80ml) ☐ Administer antibiotics (ceftriaxone 50mg/kg iv) ☐ Take history from mother	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - No push dose fluids after access, no access by 2 min → BP 75/35 <u>Triggers</u> <i>For progression to next state</i> - No glucose check by 4 min → 2. Seizure - Glucose checked, fluids given → 3. Persistent hypotension - 6 min → 3. Persistent Hypotension
2. Seizure HR → 155 BP → 145/95	Nurse states "I think she's seizing" and activates seizure. (Optional: mother to start panic "what do you mean she's seizing?")	<u>Learner Actions</u> ☐ Check glucose (2.4) , replace with 2-4ml/kg of D25 (40-80ml) ☐ Ensure staff wearing masks (meningitis risk) ☐ ± Add vancomycin for CSF penetration ☐ Delegate team member to keep mother calm and informed	<u>Modifiers</u> - Benzo given → no change to seizure <u>Triggers</u> - Glucose given → 3. Persistent Hypotension - 8 min → 3. Persistent Hypotension
3. Persistent Hypotension HR 130 BP 75/35	Patient still listless, poorly responsive.	<u>Learner Actions</u> ☐ Bolus up to total of 60ml/kg of fluid ☐ Start vasopressor (epi at 0.05 mcg/kg/min or norepi at 0.05mcg/kg/min) ☐ Consult ICU ☐ Consider intubation	<u>Modifiers</u> - 9 min (no pressor) → BP 70/30 - 10 min (no pressor) → BP 65/25 <u>Triggers</u> - Pressor started → 4. Poorly responsive - Intubation → 5. Intubation
4. Poorly Responsive HR → 120 BP → 85/45	Patient not responsive at all. BP/HR stabilized, but LOC worsening.	<u>Learner Actions</u> ☐ Consider intubation ☐ Choose correct tube size (5 uncuffed, 4.5 cuffed) ☐ Ketamine or etomidate ☐ Paralytic ☐ Apneic oxygenation	<u>Modifiers</u> - If not considering intubation → slowly decrease O ₂ SATS to 85% <u>Triggers</u> - Intubate → 5. Intubate
5. Intubation Unchanged		<u>Learner Actions</u> ☐ Intubate as above ☐ Post-intubation CXR ☐ Start sedation (midazolam) ☐ Insert OG ☐ Call ICU	<u>Modifiers</u> - Paralytics given → RR 0 <u>Triggers</u> - Intubation → END CASE - 12 min → END CASE

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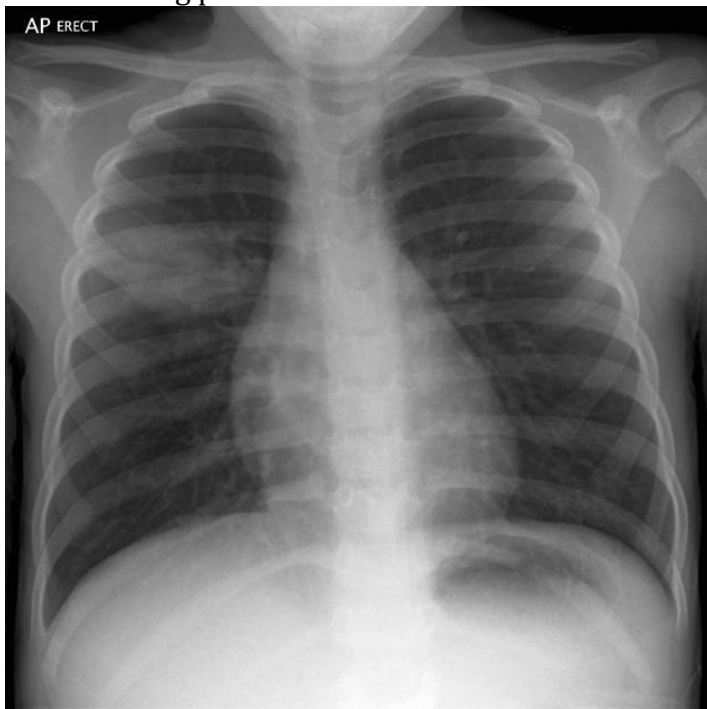
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work required for this case.

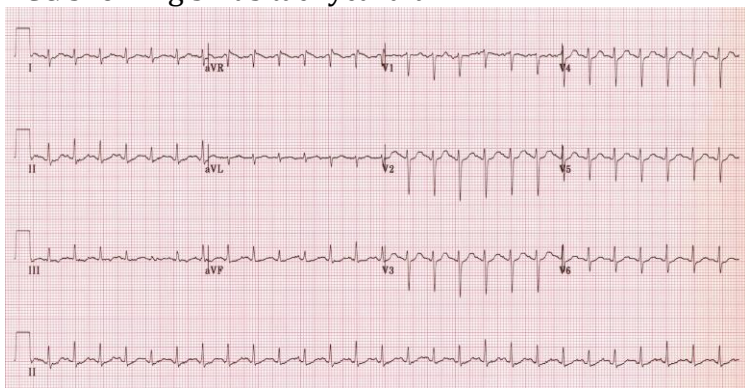
Images (ECGs, CXRs, etc.)

CXR showing pneumonia:



CXR source: <http://radiopaedia.org/articles/round-pneumonia-1>

ECG showing sinus tachycardia:



ECG source: <http://lifeinthefastlane.com/ecg-library/sinus-tachycardia/>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did it feel to perform a resuscitation with a distraught mother in the room? How do you feel the team handled the situation? Do you have any suggestions for how to improve this? 2) How did the team approach drug dosing in this child? Did you all feel comfortable with how dosing decisions were made and communicated? What are some ways to calculate weight and dosing when you are uncertain? 3) Does everyone feel comfortable putting in an IO? What are the steps? Where you can put it? 4) How do you calculate glucose replacement in a child? 5) What considerations are required for a pediatric intubation as compared to an adult intubation?	
Key Moments	
Recognition of need for IO access.	
Addressing needs of distraught mother.	
Decision to start vasopressors and intubate.	